

Assignment 2 2025

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Weighting: 30% of final course grade.

Due date/time: Sunday 28th September at 23:59.

Regarding A.I. Usage:

In this assessment task, you may use AI tools to generate materials or content related to the task. **To gain credit the AI tools must only reflect the methods taught in the course Modules.** You are expected to complete the assignment based on the examples and code covered in the lecture materials and practicals. **If you use AI to generate code, there is a high chance that the process will not align with the assignment requirements and may produce incorrect solutions. This can significantly lower your marks.** I strongly recommend that you work through the assignment independently, from start to finish, using the lecture and practical materials as your main guide. AI tools should only be used as a support to clarify concepts already covered in class—not as the primary source of code or answers. Also **always reference content generated by AI - if you don't, it's considered plagiarism.**

Tasks & Dataset files:

In this Assignment, you have two Tasks. **You will work independently on this assignment.** Your submission pdf must be in official **written report format**.

Task 1

In the first task, you will analyse and forecast the amount of horizontal solar radiation reaching the ground at a particular location over the globe. For this aim, you will work on the monthly average horizontal solar radiation and the monthly precipitation series measured at the same points between January 1960 and December 2014. Both series are available [here](#) .

Your task is to give best 2 years ahead forecasts in terms of MASE for the solar radiation series by using **all** of the time series regression methods (dLagM package), dynamic linear models (dynlm package), and exponential smoothing and corresponding state-space models covered in the Modules 3 - 7 of our course this semester. **Hint:** Use MASE() function from the dLagM package to compute MASE for time series regression methods for model comparisons.

For the solar radiation forecasts, the required precipitation measurements (predictor series) for the months from January 2015 to December 2016 at the exact same locations are given in [data.x.csv](#)  file. You will use this data for the calculation of 2 years ahead forecasts.

Note: You must complete all elements listed in the rubric when working on the task. To receive credit, you **must use methods or functions covered in Modules 3 to 7**. Please be aware that **if you use any method not covered in the module content, you will not receive credit** for that part.

Task 2

In the second task, you will analyse the correlation between quarterly Residential Property Price Index (PPI) in Melbourne and quarterly population change over the previous quarter in Victoria between September 2003 and December 2016. The quarterly PPI and population change series are available in this [data2.csv](#)  file.

The main goal of your analysis is to **demonstrate** whether the correlation between these two series is spurious or not.

Note: You must complete all elements listed in the rubric when working on the task. To receive credit, you **must use methods or functions covered in course Modules**. Please be aware that **if you use any method not covered in the course content, you will not receive credit** for that part.

Summary

In both tasks, it is expected that you ask yourself what are the elements of a suitable and successful data analysis, and how you might go about presenting your results in a **written report format**. Please review the contents of the relevant modules and apply suitable approaches here. The **rubric** given below will guide you through my expectations in terms of reporting, R codes, descriptive analysis, modelling, and diagnostic checking.

Note: You must complete all elements listed in the rubric when working on the tasks. To receive credit, you **must use methods or functions covered in course Modules**. Please be aware that **if you use any method not covered in the course content, you will not receive credit** for that part.

Written report format:

Please see the following webpage to get more information about the required **written report format**:

<https://learninglab.rmit.edu.au/assessments/reports/> ➞

Choice regarding Microsoft Word or R Markdown:

In regards to creating your pdf submission for this and later Assignments, you are free to simply write your R code, run it, save your output, and then combine both R code and output with your interpretations in a Word document, and then you must save that as a **single pdf file**.

Or alternatively, you can invest time into learning R Markdown coding, but you will need to save your R Markdown output as a pdf (do not submit a .Rmd file in Canvas). Many of you are familiar with R Markdown, but for those of you who are not, here are official R Markdown learning resources if you are interested:

<https://rmarkdown.rstudio.com/> ➞

<https://rmarkdown.rstudio.com/lesson-1.html> ➞

Submission Instructions:

- All reports must be **submitted via the Canvas upload links on this assignment page**.
- Your submission should be **a SINGLE PDF file only**.
- Your report should meet with English language requirements.
- **Late submissions** will be penalised in accordance with the late submission policy explained in the "Course Overview" module pages.

Plagiarism and Academic Misconduct:

Please keep in mind that this is an individual assignment and you should demonstrate your own effort and understanding. If cheating is detected, both the copier and the student copied from will be responsible. Therefore, it is your responsibility to ensure you do not copy or do not allow other classmates to copy your work. Please read through the [academic integrity pages of RMIT University](#) ➞. It is important to understand proper referencing the information that you use in your writings. Be aware that plagiarism and academic misconduct are serious offenses. Please review the [RMIT University policy](#) ➞ on plagiarism.

Marking Rubric:

Dot-pointed below are the main components of your assignment that will be marked. Further details on component sub-grades can be found in the detailed Rubric table at the bottom of the page.

- Reporting & Interpretation (20%)
- R codes (25%)

- Descriptive Analysis (10%)
- Choice of variables/model (15%)
- Implementation of models including Forecasting (15%)
- Diagnostic checking (15%)

Criteria	Ratings							Pts
Reporting & Interpretation	20 Pts >16 to 20 Achieves High distinction All inferences are clearly addressed by relevant outputs and given under relevant sections. All bits of the presented output are used for the inferences in a way to provide a mature discussion of analysis results. There is no missing section. There are no language and formatting issues.	16 Pts >14 to 16 Achieves Distinction All inferences are clearly addressed by relevant outputs and given under relevant sections. All bits of the presented output are used for the inferences. There is no missing section. There are no language and formatting issues.	14 Pts >12 to 14 Meets expectation All inferences are addressed by relevant outputs and given under relevant sections. There is no missing section. There are no language and formatting issues.	12 Pts >8 to 12 Pass All inferences are addressed by relevant outputs and given under relevant sections. There is no missing section. There are no language and formatting issues.	8 Pts >4 to 8 Needs improvement Inferences are addressed by relevant outputs and given under relevant sections. There are missing sections. There are minor languages and formatting issues.	4 Pts >0 to 4 Poor Inferences are not addressed by relevant outputs and given under relevant sections. Some of the output remains without being inferred. There are language and formatting issues.	0 Pts No marks Lacks the basic elements of a precise report. Some of the outputs have not been associated with inferences or there are inferences without supporting evidence.	20 pts
R codes	25 Pts >20 to 25 Achieves High distinction Scripts to run analyses are working properly and suitable explanations are given on the code. Suitable object names are used to make it easy to follow the code. The scripts include user-created functions to implement all analyses in a	20 Pts >18 to 20 Achieves Distinction Scripts to run analyses are working properly and suitable explanations are given on the code. Suitable object names are used to make it easy to follow the code. The scripts include user-created functions to implement all analyses without using	18 Pts >15 to 18 Meets expectation Scripts to run analyses are working properly and suitable explanations are given on the code. Suitable object names are used to make it easy to follow the code. The script is arranged in a way not to repeat the same code chunks.	15 Pts >10 to 15 Pass Scripts to run analyses are working properly and suitable explanations are given on the code. Suitable object names are used to make it easy to follow the code.	10 Pts >5 to 10 Needs improvement All scripts to run analysis are working properly and suitable explanations are given on the code.	5 Pts >0 to 5 Poor All scripts to run analyses are working properly. However, it is hard to follow codes to identify what each line does.	0 Pts No marks Some of the scripts to run analyses are generating error messages. It is unclear what each line does in the R codes.	25 pts

Criteria	Ratings							Pts
	dynamical way.	the same code chunks repeatedly.						
Descriptive analysis	10 Pts >8 to 10 Achieves High distinction Uses all suitable plots and descriptive statistics are displayed. All plots and descriptive statistics are interpreted in a way to demonstrate an outstanding understanding of concepts targeted by the descriptive statistic/plot.	8 Pts >7 to 8 Achieves Distinction Uses all suitable plots and descriptive statistics are displayed. All plots and descriptive statistics are interpreted in a way to shed light on further analyses.	7 Pts >6 to 7 Meets expectation Uses all suitable plots and descriptive statistics and each plot and the descriptive statistic have a suitable interpretation.	6 Pts >4 to 6 Pass Uses all suitable plots and descriptive statistics and some plots and descriptive statistic have a suitable interpretation in one of the tasks.	4 Pts >2 to 4 Needs improvement Some descriptive plots/statistics are used and/or presented all plots/statistics have interpretations.	2 Pts >0 to 2 Poor Some descriptive plots/statistics are used and/or presented plots/statistics have interpretations.	0 Pts No marks Nearly no descriptive analysis has been done and/or the descriptive analysis is insufficient.	10 pts
Choice of variables/model	15 Pts >12 to 15 Achieves High distinction A suitable model is chosen, and all variables are suitable for the aim of analysis. Insightful reasons given about the selection of the model.	12 Pts >11 to 12 Achieves Distinction A suitable model is chosen, and all variables are suitable for the aim of analysis. The reason behind the selection of the model is articulated based on the results of descriptive statistics.	11 Pts >9 to 11 Meets expectation A suitable model is chosen, and all variables are suitable for the aim of analysis. The reason behind the selection of the model is explained.	9 Pts >6 to 9 Pass A suitable model is chosen, and all variables are suitable for the aim of analysis.	6 Pts >3 to 6 Needs improvement A suitable model is chosen but some variables in a model are unsuitable for the aim of analysis.	3 Pts >0 to 3 Poor A suitable model is chosen but some or all variables in a model are unsuitable for the aim of analysis.	0 Pts No marks An unsuitable model is chosen or all variables in a model are unsuitable for the aim of analysis.	15 pts
Implementation of models including Forecasting	15 Pts >12 to 15 Achieves High distinction All possible models are fitted, and all the relevant outputs are clearly	12 Pts >11 to 12 Achieves Distinction All possible models are fitted, and all the relevant outputs are included in the	11 Pts >9 to 11 Meets expectation All possible models are fitted and there is some effort to draw inferences	9 Pts >6 to 9 Pass All possible models are fitted but there is a limited amount of effort to draw inference from the	6 Pts >3 to 6 Needs improvement Some models fitted but there is a limited amount of effort to draw inference	3 Pts >0 to 3 Poor Some models fitted but there is no attempt to draw inferences from the	0 Pts No marks Specified models are not fitted and there is no attempt to move forward with the analysis.	15 pts

Criteria	Ratings							Pts
	interpreted in a way to inform the next step of the analysis.	comments made on the fitted models.	from the model outputs.	model outputs.	from the model outputs.	model outputs.		
Diagnostic checking	15 Pts >12 to 15 Achieves High distinction All aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated. A mature discussion of the information gathered by diagnostic checking is given.	12 Pts >11 to 12 Achieves Distinction All aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated. The results are articulated in a way to demonstrate a good understanding of diagnostic checking.	11 Pts >9 to 11 Meets expectation All aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated.	9 Pts >6 to 9 Pass Some aspects of diagnostic checking are applied, and all assumptions of applied approaches are validated.	6 Pts >3 to 6 Needs improvement Some aspects of diagnostic checking are applied and most of the assumptions of applied approaches are validated.	3 Pts >0 to 3 Poor Some aspects of diagnostic checking are applied and some of the assumptions of applied approaches are validated.	0 Pts No marks There is no effort for diagnostic checking and validation of the assumptions of applied approaches.	15 pts
Total points: 100								